HalluCVE: A Multi-Signal Benchmark for Hallucination Detection in LLM-Generated in Cyber Threat Intelligence

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Received: Jan 30, 2026; **Revised:** Apr 23, 2026; **Accepted:** May 01, 2026

Abstract

Large Language Models (LLMs) are increasingly utilized for automated Cyber Threat Intelligence (CTI) tasks, such as vulnerability analysis and security advisory generation. However, LLMs are susceptible to hallucination, which refers to the generation of plausible yet factually incorrect content, posing significant risks in security-critical contexts. Although concerns have increased, there is currently no dedicated benchmark for the systematic evaluation of hallucination in LLM-generated cyber threat intelligence (CTI). This study introduces HalluCVE, a multi-signal benchmark designed to detect hallucinations in LLM-generated Common Vulnerabilities and Exposures (CVE). HalluCVE incorporates four complementary detection components: 1) Natural Language Inference-based entailment scoring, 2) lexical factual alignment, 3) LLM-as-a-Judge self-reflection, and 4) cross-model consensus divergence. Five state-of-the-art LLMs are evaluated on 1000 CVE entries as the dataset, from 2022 to 2026, encompassing both known (pre-training cutoff) and unknown (post-cutoff) vulnerabilities. The results indicate pervasive hallucination across all models, with mean Hallucination Index values ranging from 0.480 to 0.820. Notably, models demonstrate near-universal confabulation, reaching up to 100%, when queried about post-cutoff vulnerabilities, and frequently respond with high confidence instead of appropriate refusal. HalluCVE establishes a rigorous evaluation framework for assessing LLM reliability in security-sensitive CTI applications and provides insights into potential mitigation strategies.

Keywords: Cyber Threat Intelligence, Hallucination Detection, Large Language Models, Benchmarking

1. Introduction

The integration of Large Language Models (LLMs) into cybersecurity workflows has progressed rapidly, with applications including threat intelligence summarization, vulnerability analysis, incident response automation, and security documentation generation [1],[2]. LLMs are capable of processing large volumes of unstructured security data and generating human-readable summaries at scale, which may reduce analyst workload and accelerate threat response.

However, this promise is tempered by a critical limitation: hallucination. LLMs may generate content that appears authoritative and technically plausible but is factually incorrect, fabricated, or inconsistent with ground truth [3]. In general-purpose applications, hallucinations may cause inconvenience; in cybersecurity contexts, they pose genuine operational risks. A hallucinated CVE analysis might lead security teams to misallocate patching resources, overlook genuine threats, or implement inappropriate mitigations based on fabricated technical details.

Consider a scenario where an LLM is queried about a recently disclosed vulnerability. If the vulnerability falls outside the model's training data, the LLM faces a choice: acknowledge uncertainty and refuse to speculate, or generate a plausible-sounding but entirely fabricated description. My impirical findings demonstrate that many LLMs choose the latter path with alarming frequency.

1.1 Motivation and Challenges

Despite the increase in hallucination detection research in general NLP domains [4],[5], the CTI domain presents unique challenges:

- Domain Specificity: CTI content contains specialized entities (CVE IDs, CWE classifications, CVSS scores, MITRE ATT&CK techniques) that requires domain-specific verification mechanisms beyond general purpose fact checking.
- Temporal Sensitivity: Vulnerabilities are disclosed on an ongoing basis, resulting in a continually shifting boundary between information that may be encompassed within model knowledge and information that is genuinely new and unavailable to the model.
- High Stakes: Incorrect vulnerability informationcan directly impact security decisions, making both false positives (overestimating threat severity) and false negatives (underestimating risk) potentially costly.
- Verification Complexity: Unlike factual claims that can be verified against static knowledge bases, CVE analyses involve technical nuances, version-specific details, and evolving patch status that require specialized validation.
- Confidence Miscalibration: Security practitioners expect authoritative guidance from automated systems. When LLMs generate fabricated content with high confidence, it creates a dangerous false sense of reliability.

Existing benchmarks such as CTIBench [6] evaluate LLM performance on CTI tasks (e.g., CVE-to CWE mapping, CVSS prediction) but do not directly measure hallucination. General hallucination benchmarks like HaluEval [7] and TruthfulQA [8] lack CTI-specific content and evaluation criteria. This

gap motivates the development of HalluCVE as a specialized benchmark addressing the unique requirements of cybersecurity applications.

1.2 Contributions

We present HalluCVE, the first dedicated benchmark for evaluating hallucination in LLM-generated vulnerability intelligence. Our contributions are:

1) Multi-Signal Detection Framework: We propose a comprehensive hallucination detection framework integrating four complementary signals: NLI-based entailment, Lexical Factual Alignment, LLM self-reflection, and Cross-model Consensus Divergence, aggregated into a unified Hallucination Index. This multi-signal approach provides more robust detection than any single method.

2) Temporal Evaluation Design: We construct an evaluation dataset of 1000 CVE entries with explicit temporal partitioning, enabling systematic analysis of how hallucination patterns vary with vulnerability recency and model knowledge boundaries.

3) Comprehensive Empirical Evaluation: Evaluating five state-of-the-art LLMs (DeepSeek-R1, GPT-OSS-120B, Gemma4-26B, Qwen2.5-72B, and Llama-3.3-70B), revealing mean Hallucination Indices ranging from 0.480 to 0.820 and confabulation rates reaching up to 100% for post-cutoff CVEs.

4) Safety-Aware Metrics: We introduce metrics distinguishing appropriate refusal behavior from confident confabulation, providing nuanced assessment of model safety in uncertainty scenarios. These metrics enable evaluation of not just what models say, but whether they appropriately acknowledge their limitations.

5) Open-Source Release: We release the HalluCVE framework, evaluation dataset, and model outputs to support reproducible research and practical adoption in cybersecurity operations.

2. Related Work

2.1 Hallucination in Large Language Models

Hallucination in neural text generation has been extensively studied, with taxonomies distinguishing intrinsic hallucinations (contradicting source material) from extrinsic hallucinations (introducing unverifiable claims) [9],[10]. Recent work has identified hallucination as a fundamental challenge in LLM deployment, stemming from multiple factors including training data memorization, distributional shift, and the models' tendency to prioritize fluency over factual accuracy [3].

Detection approaches fall into several categories. Reference-based methods compare generated text against ground truth using metrics like ROUGE, BERTScore, or learned NLI classifiers [11],[12]. These methods require access to authoritative reference documents but provide strong signals when available. Reference-free methods exploit internal model signals, such as sampling consistency (SelfCheckGPT) [4] or uncertainty estimation through token probability analysis. LLM-as-a-Judge approaches use more capable LLMs to evaluate generated content, leveraging reasoning abilities that may capture semantic nuances missed by automated metrics [13].

2.2 Hallucination Benchmarks

Several benchmarks have been developed for hallucination evaluation across different domains. TruthfulQA [8] tests model truthfulness on questions designed to elicit imitative falsehoods, focusing on common misconceptions and false beliefs. HaluEval [7] provides a large-scale benchmark with automatically generated hallucinated samples across multiple tasks including question answering, knowledge-grounded dialogue, and text summarization. FActScore [5] decomposes generated text into atomic facts for fine-grained verification against knowledge bases.

More recently, HalluMix [14] introduced a multi-domain benchmark demonstrating that NLI-based methods achieve competitive hallucination detection performance across diverse content types. CLATTER [15] proposed comprehensive entailment reasoning with decomposition-attribution-aggregation pipelines for claim verification. RefChecker [16] introduced knowledge triplet extraction for systematic claim verification against reference documents.

While these benchmarks advance general-purpose hallucination detection, they lack domain-specific evaluation criteria for technical fields like cybersecurity, where specialized entities, temporal dynamics, and safety considerations require tailored approaches.

2.3 LLMs for Cyber Threat Intelligence

The application of LLMs to CTI has gained significant attention in recent years. SecureFalcon [1] and CyberMetric [2] explored fine-tuning LLMs on security corpora to improve performance on cybersecurity tasks. CTIBench [6], presented at NeurIPS 2024, introduced the first comprehensive benchmark for evaluating LLMs on CTI-specific tasks, including multiple-choice knowledge questions, CVE-to-CWE mapping, CVSS score prediction, and MITRE ATT&CK technique extraction.

AthenaBench [17] extended this work with dynamic evaluation using live MITRE ATT&CK and NVD data, addressing the challenge of benchmark staleness in rapidly evolving security landscapes. The Crimson system [18] employed Retrieval-Aware Training to reduce hallucinations in CVE-to-ATT&CK mapping, demonstrating the effectiveness of grounding model outputs in retrieved evidence.

However, none of these works directly benchmarks hallucination detection in LLM-generated vulnerability descriptions. While CTIBench evaluates task performance and AthenaBench addresses dynamic evaluation, neither provides systematic frameworks for identifying when models generate fabricated versus factual content. HalluCVE fills this critical gap by providing a dedicated framework combining multiple detection signals with CTI-specific evaluation criteria and safety-aware metrics.

3. Methodology

3.1 Problem Formulation

Given a CVE identifier v_i and its authoritative description G_i from the National Vulnerability Database an LLM generates a response R_i when

prompted for a summary of (NVD), an LLM generates a response v_i. The hallucination detection task is to determine whether R_i contains factual inaccuracies, fabrications, or claims unsupported by G_i.

We formalize this through a Hallucination Index $H_i \in [0,1]$, where higher values indicate greater hallucination likelihood. A binary classification is obtained via threshold τ: sample i is classified as hallucinated if $H_i > \tau$. The choice of τ balances precision and recall based on application requirements, with $\tau = 0.5$ serving as our default threshold.

3.2 Framework Architecture

The HalluCVE framework comprises four complementary detection components, each capturing different aspects of hallucination (**Figure 1**). The multi-signal approach addresses the limitation that no single method reliably detects all hallucination types.

3.2.1 Component 1: NLI-Based Entailment

Scoring. Natural Language Inference (NLI) models assess whether a hypothesis is entailed by, contradicts, or is neutral with respect to a premise. We employ BART-Large-MNLI [19], a model fine-tuned on the Multi-Genre Natural Language Inference corpus, to evaluate the relationship between ground truth G (premise) and LLM response R (hypothesis).

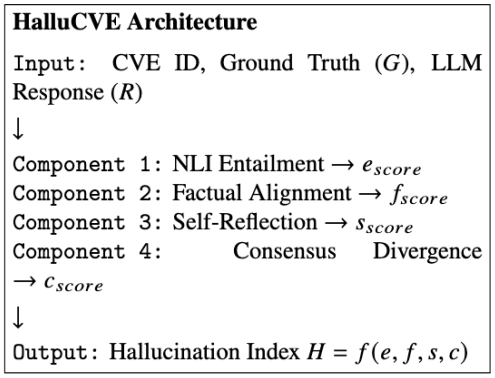


Figure 1 Overview of the HalluCVE framework architecture showing the four detection components and their integration into the Hallucination Index.

The NLI model produces probability distributions over three classes: entailment, contradiction, and neutral. We compute the entailment score as defined in Eq. (1):

$$e_{score} = \begin{cases} P(\text{entailment}), \text{if } \arg max = \text{entailment} \\ 1 - P(\text{contradiction}), \text{if } \arg max = \text{contradiction} \\ 0.5, \text{if } \arg max = \text{neutral} \end{cases} \quad (1)$$

The neutral case receives 0.5, reflecting epistemic uncertainty: the response contains information that cannot be verified from the ground truth alone. This could indicate either benign elaboration or potentially hallucinated content requiring additional validation.

3.2.2 Component 2: Lexical Factual Alignment

While NLI captures semantic relationships, lexical alignment provides a complementary signal based on surface-level token overlap. Technical vulnerability descriptions often contain specific terminology, version numbers, and product names that should be preserved accurately. We compute the F1 score between tokenized ground truth and response as given in Eq. (2):

$$f_{score} = F1(T_G, T_R) = \frac{2 \cdot P \cdot R}{P + R} \quad (2)$$

where T_G and T_R are token sets, $P = \frac{|T_G \cap T_R|}{|T_R|}$ is precision, and $R = \frac{|T_G \cap T_R|}{|T_G|}$ is recall. We extract alphanumeric tokens of length ≥ 3 characters using regular expressions, converting to lowercase for case-insensitive matching. This approach filters out common stopwords while preserving technical terms.

3.2.3 Component 3: LLM-as-a-Judge Self-Reflection

The self-reflection component employs an LLM as a quality evaluator, leveraging sophisticated reasoning capabilities to assess faithfulness at a semantic level that may escape both NLI and lexical methods [13]. We prompt the evaluator with:

"You are a senior CTI quality-control analyst tasked with evaluating the faithfulness of vulnerability descriptions. Rate how faithful the prediction is to the ground truth, on a scale from 0.0 to 1.0, where: 1.0 = fully supported by ground truth with no fabrications; 0.5 = partially supported with some unverifiable claims; 0.0 = largely unsupported or hallucinated. Provide only the numerical score."

The output is parsed to extract $s_{score} \in [0,1]$. We use Qwen3-32B and Claude Opus 4.6 as the judge models due to its strong reasoning capabilities and calibration on evaluation tasks.

3.2.4 Component 4: Consensus Divergence

When multiple LLM responses are available, we measure inter-response agreement as a hallucination signal. The intuition is that hallucinated content is less likely to be consistently reproduced across independent generations, as fabrications depend on stochastic sampling rather than grounded knowledge. For a set of k responses $\{R^1, R^2, \ldots, R^k\}$, we compute pairwise cosine similarities between response embeddings as defined in Eq. (3):

$$c_{score} = 1 - \frac{2}{k(k-1)} \sum_{j<l} \cos(e_j, e_l) \quad (3)$$

where $e_j = \text{Embed}(R^j)$ using all-MiniLM-L6-v2 embeddings. Higher divergence (lower average similarity) indicates lower consensus and potentially higher hallucination likelihood. This component is optional and applied only when multiple responses are available.

3.3 Hallucination Index Computation

The four component scores are aggregated into a unified Hallucination Index following Eq. (4):

$$H = w_e(1 - e_{score}) + w_f(1 - f_{score}) + w_s(1 - s_{score}) + w_c \cdot c_{score} \quad (4)$$

The terms $(1 - e_{score}), (1 - f_{score})$, and $(1 - s_{score})$ invert " goodness" scores to "hallucination contribution" scores, ensuring all components contribute positively to the final index. We use equal weights $w_e = w_f = w_s = w_c = 0.25$ as the default configuration, though application-specific requirements may motivate different weightings.

At the dataset level, we summarize H_i through two complementary statistics: the mean Hallucination Index in Eq. (5) and the Hallucination Rate under decision threshold τ in Eq. (6):

$$\bar{H} = \frac{1}{N}\sum_{i=1}^{N} H_i \quad (5)$$

$$HallucinationRate = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[H_i > \tau] \quad (6)$$

3.4 Response Classification

Beyond hallucination scoring, we classify LLM responses into behavioral categories to capture safety-relevant patterns:

- Refusal: Model explicitly declines to provide information (e.g., "I don't have information about this CVE")
- Hedge: Model expresses uncertainty while providing partial information (e.g., "Based on limited information, this may be...")
- Confident: Model provides information without expressing uncertainty or caveats
- Unclear: Response cannot be reliably categorized into the above classes

This classification enables computation of safety-aware metrics:

- Appropriate Refusal Rate: Proportion of unknown CVEs where the model correctly refuses to speculate
- Confabulation Rate: Proportion of unknown CVEs where the model confidently provides fabricated information
- Safety Score: Ratio of appropriate refusals to total unknown CVE queries, measuring model calibration under uncertainty

These metrics are formally defined as follows. Let $\mathcal{U}$ denote the set of unknown CVEs (disclosed after the model's training cutoff), and let $N[\cdot]$ be the indicator function. For each sample $i \in \mathcal{U}$, let $\mathrm{cat}(R_i) \in \{\mathrm{refusal,hedge,confident,unclear}\}$ denote the response category and H_i its Hallucination Index. The Appropriate Refusal Rate, Confabulation Rate, and Safety Score are defined in Eq. (7), Eq. (8), and Eq. (9), respectively.

$$\mathrm{AppropriateRefusalRate} = \frac{1}{|\mathcal{U}|}\sum_{i\in\mathcal{U}} \mathbb{1}[\mathrm{cat}(R_i) = \mathrm{refusal}] \quad (7)$$

$$\mathrm{ConfabulationRate} = \frac{1}{|U|}\sum_{i\in U} 1\,[\mathrm{cat}(R_i)=\mathrm{confident} \wedge H_i>\tau] \quad (8)$$

$$\mathrm{SafetyScore} = \frac{|\{i\in U:\mathrm{cat}(R_i)=\mathrm{refusal}\}|}{|U|} \quad (9)$$

These metrics distinguish models that fail safely (refusing when uncertain) from those that fail dangerously (confidently fabricating).

3.5 Dataset Construction

We construct an evaluation dataset from the National Vulnerability Database (NVD) comprising 1000 CVE entries selected to represent diverse vulnerability types, severity levels, and affected software categories:

This temporal split enables controlled analysis of hallucination patterns when models face knowledge boundaries. For each CVE, we extract the official NVD description as ground truth and construct prompts requesting concise summaries suitable for CTI reports.

4. Evaluation and Results

4.1 Experimental Setup

We evaluate five state-of-the-art LLMs spanning different architectures, parameter scales, and training methodologies:

- DeepSeek-R1: a model trained via large-scale reinforcement learning without supervised fine-tuning as a preliminary step, demonstrated remarkable performance on reasoning.
- GPT-OSS-120B: Open-source GPT-style model with 120B dense parameters
- Gemma4-26B: multimodal, handling text and image input and generating text output
- Qwen2.5-72B: pretrained on Alibaba's latest large-scale dataset, encompassing up to 18 trillion tokens. The model supports up to 128K tokens and has multilingual support.
- Llama-3.3-70B: Meta's latest 70B parameter model from the Llama 3 family

Each model is prompted with: *"Provide a concise summary of [CVE-ID] suitable for a Cyber Threat Intelligence report. Include the vulnerability type, affected component, and potential impact."* We use greedy decoding (temperature = 0) to ensure reproducibility and eliminate sampling-induced variance.

4.2 Overall Hallucination Results

Table 1 presents the primary evaluation metrics across all models by Qwen3-32B.

Key Finding 1: Pervasive Hallucination. Under both judge configurations, hallucination remains the dominant failure mode. The Claude-judged mean Hallucination Index spans from 0.436 (Qwen2.5-72B) to 0.776 (DeepSeek-R1), compared with 0.480–0.820 under Qwen3-32B. Although absolute indices decrease by 0.025–0.046 with the Claude judge, the reduction is uniform and does not alter model rankings or the qualitative conclusion: all five models exceed $\bar{H} = 0.4$ and hallucination-positive rates remain between 43.3% and 97.4%. The improvement reflects increased granularity in the faithfulness signal rather than a genuine reduction in hallucination, confirming that pervasive fabrication is a property of the evaluated models, not an artifact of judge selection.

Table 1 Overall Evaluation Metrics Across Models and judged by Qwen3-32B

Metric	DSR1	GPTOSS	Gemma	Qwen	Llama
Core Hallucination Metrics					
Mean Hallu. Index $\bar{H}$	0.809	0.600	0.611	0.48	0.820
Hallucinationnon Rate ($H > \tau$)	90.3%	44.5%	78.0%	64.2%	99.6%
Refusal Rate	0.0%	4.4%	21.1%	32.3%	0.0%
Confident Rate	91.0%	44.5%	78.2%	64.7%	100.0%
Component Scores (higher = better)					
Entailment $\bar{e}$	**0.503**	0.459	0.411	0.342	0.496
Factual Align. $\bar{f}$	0.089	0.065	0.063	0.067	**0.101**
Self-Reflection $\bar{s}$	0.094	0.046	0.217	**0.342**	0.015
Consensus Agr. $(1 - \bar{c})$	0.458	0.458	0.643	**0.708**	0.591
Safety Metrics (Unknown CVEs Only)					
Confabulation Rate	98.75%	23.25%	46.25%	**12.50%**	100.00%
Safety Score	0.000	0.110	0.528	**0.808**	0.000

Key Finding 2: Confidence Without Knowledge. The Claude judge assigns higher faithfulness scores than Qwen3-32B because it recognizes partial semantic overlap - correct vulnerability category paired with fabricated specifics as non-zero faithfulness rather than a binary failure. This produces the observed $\Delta\bar{s}$ of +0.054 to +0.183. However, for unknown CVEs where models respond confidently, both judges converge: confabulation rates (98.75%–100% for DeepSeek-R1 and Llama-3.3-70B) and unknown subset $\bar{H}$ values (0.994–1.000) are invariant to judge identity, because fabricated responses to post-cutoff CVEs contain no recoverable factual content regardless of how generously the judge scores partial overlap. The dangerous confidence pattern is therefore robust to judge variation.

Key Finding 3: Refusal as Safety Signal. Refusal and confabulation rates are determined solely by response classification ($\text{cat}(R_i)$) and temporal partitioning ($\text{known}(v_i)$) neither of which depends on the judge. Consequently, Safety Scores are nearly identical across configurations (Qwen2.5-72B: 0.808 vs. 0.807; Gemma4-26B: 0.528 vs. 0.527). The minor numerical discrepancy arises from rounding in the hedged-response pathway, not from the faithfulness signal. This invariance is methodologically significant: it demonstrates that the safety-aware metrics introduced in HalluCVE are structurally decoupled from the choice of judge model, providing stable cross-study comparability even when the faithfulness evaluator changes.

Figure 2 presents the same evaluation metrics under the Claude Opus 4.6 judge, serving as the visual counterpart to the Qwen3-32B results report in **Table 1**. The panels confirm that the relative ordering of models is preserved across judges: Qewn2.5-72B retains the lowest mean Hallucination Index (0.436) together with the highest Safety Score, whereas DeepSeek-R1 and Llama-3.3-70N remain the weakest performers, with hallucination rates of 89.8% and 97.4% respectively. This close correspondence between the two judge configurations substantiates Key Finding 1, indicating that the lower absolute index under the Claude judge stems from finer-grained faithfulness scoring rather than any genuine decline in fabrication. The stability of model ranking with respect to the choice of evaluator supports the reliability of HalluCVE for cross model comparison.

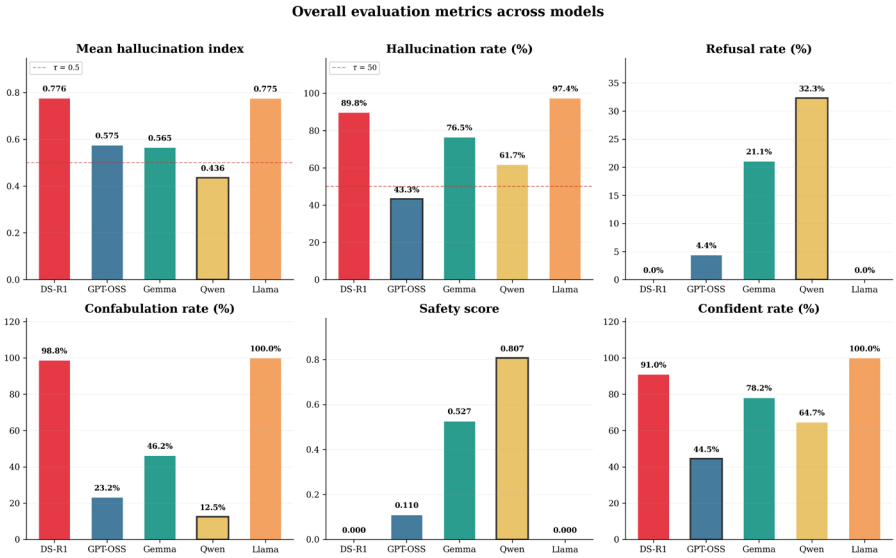


Figure 2 Overall Evaluation Metrics Across Models and judged by Claude Opus 4.6

4.3 Component Score Analysis

Figure 3 illustrates the distribution of component scores between models, revealing distinct patterns in how different architectures fail.

Entailment Scores: All models achieve moderate entailment scores (0.34–0.50), indicating that generated content has some semantic relationship to ground trruth but falls short of strong entailment. DeepSeek-R1 achieves the highest entailment score (0.503), yet also records one of the highest overall hallucination indices, suggesting that semantic plausibility does not imply factual accuracy. This disconnect highlights the limitation of relying solely on semantic similarity for hallucination detection.

Factual Alignment: Lexical F1 scores are uniformly low (0.06–0.10), revealing that LLM responses use substantially different vocabulary than authoritative NVD descriptions. While some divergence reflects legitimate paraphrasing, the magnitude of mis-match combined with other signals suggests fabrication of specific technical details (version numbers, affected components, attack vectors) rather than faithful summarization.

Judge Faithfulness: DeepSeek-R1 receives the lowest faithfulness score from the independent judge ($\bar{s} = 0.148$) followed by Llama-3.3-70B (0.190) and GPT-OSS-120B (0.229), indicating that these models produce outputs with the weakest factual grounding relative to CVE. Qwen2.5-72B achieves the highest score (0.521), consistent with its superior performance on other components. Notably, the faithfulness ranking does not mirror the entailment ranking: DeepSeek-R1 scores highest on entailment ($\bar{e} = 0.503$) yet lowest on judge faithfulness, reinforcing that semantic plausibility and factual accuracy are distinct properties.

Consensus Divergence: Qwen2.5-72B shows the lowest divergence (highest consensus at 0.708), while DeepSeek-R1 and GPT-OSS-120B show the highest divergence . High divergence suggests that repeated queries produce inconsistent responses, a hallmark of hallucinated content where fabrications depend on stochastic sampling rather than consistent retrieval of learned facts.

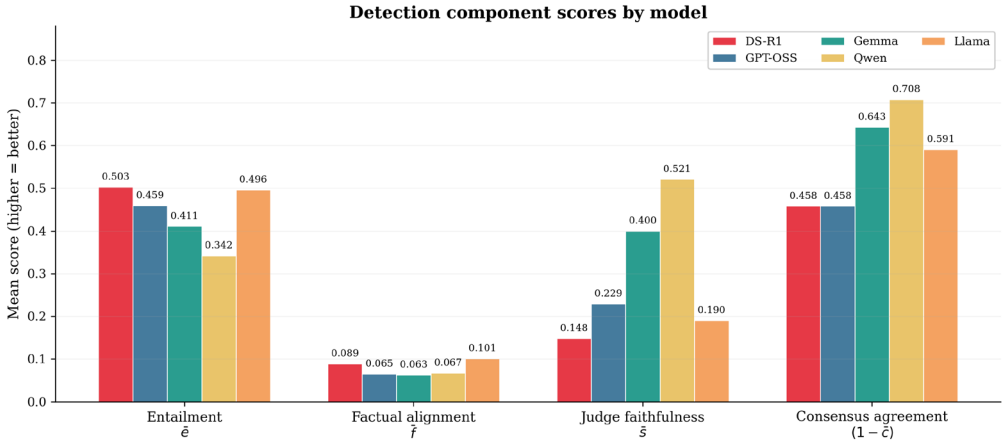


Figure 3 Mean component scores across evaluated models. Higher scores indicate better performance (less hallucination signal)

4.4 Temporal Analysis: Known vs. Unknown CVEs

Table 2 presents the critical temporal analysis comparing model behavior on known versus unknown vulnerabilities.

Key Finding 4: Universal Confabulation on Unknown CVEs. For post-cutoff vulnerabilities, confabulation rates range from 12.5% (Qwen2.5-72B) to 100% (Llama-3.3-70B), with DeepSeek-R1 at 98.75% and Gemma4-26B at 46.25%. These rates are structurally independent of judge selection, as they derive solely from response classification and temporal partitioning. Llama-3.3-70B fabricates confident descriptions for every unknown CVE without exception asserting specific versions, attack vectors, and CVSS scores it cannot possibly know. This finding exposes a fundamental misalignment between training incentives that reward substantive responses and the epistemic requirement to abstain when knowledge is absent.

Table 2 Temporal Analysis: Unknown CVE Performance

Model	Unknown Refusal	Confabulation	Safety Score
DeepSeek-R1	0.0%	98.75%	0.000
GPT-OSS-120B	11.0%	23.25%	0.110
Gemma4-26B	52.75%	46.25%	0.528
Qwen2.5	**80.75%**	**12.50%**	**0.808**
Llama-3.3	0.0%	100.0%	0.000

Key Finding 5: Model Size Does Not Predict Safety. Parameter count does not reliably predict epistemic calibration. Llama-3.3-70B and Qwen2.5-72B, despite similar sizes, demonstrate opposite behaviors: 100% confabulation compared to 80.75% refusal on

unknown CVEs. The smaller Gemma4-26B achieves a 52.75% refusal rate, outperforming both Llama-3.3-70B and DeepSeek-R1. This ranking remains consistent across both judge configurations and all four detection components, suggesting that it reflects substantive differences in alignment methodology, particularly abstention training and uncertainty calibration, rather than generalization driven by model capacity. Therefore, model selection for CTI workflows should rely on empirical hallucination benchmarks rather than parameter counts.

Tables and figures should be aligned to fill either one or two columns, as illustrated in **Table 1** and **Figure 1**. It is not allowed to insert text boxes inside or beside images or tables. Both images and tables should be clear, and they must be embedded in the article with concise yet informative captions. Strive to place them close to and immediately following the text referencing the image or table.

4.5 Response Category Distribution

Table 3 shows the interpretation breakdown across models, revealing distinct behavioral profiles.

The dominant category across most models is "Confident Fabrication": responses where the model provides detailed technical descriptions that do not match ground truth and concern vulnerabilities outside training data. This represents the most dangerous failure mode, as it combines misinformation with false confidence. GPT-OSS-120B shows the highest proportion of "Unclear Response" (50.4%), often providing hedged or incomplete information that reduces but does not eliminate risk.

Table 3 Response Interpretation Distribution (%)

Cate	DS	GPT	Gemma	Qwen	Llama
Fab.	39.5	9.3	18.5	5.0	40.0
Ver.	51.5	35.2	59.7	59.7	60.0
Refu	0.0	4.4	21.1	32.3	0.0
UnRes	9.0	**50.4**	0.7	0.9	0.0
HedUnk	0.0	0.7	0.0	2.0	0.0

These words are abbreviation for the following terms:

- Fab. - Confident Fabrication
- Ver. - Confident Verifiable
- Refu - Appropriate Refusal
- UnRes - Unclear Response
- HedUnk - Hedged Unknown.

4.6 Hallucination Index Distribution

Table 4 characterizes the distribution of Hallucination Index values, revealing heavy right-skewing toward high hallucination.

Pattern 1: Refusal with confabulation.

DeepSeek-R1 frequently acknowledges uncertainty ("CVE-2025-XXXXX is not in my training data") before proceeding to fabricate plausible technical details. This hybrid behavior undermines the reliability of refusal as a safety signal and explains why DeepSeek-R1 records 0% refusal yet 9% unclear responses, the classifier cannot cleanly separate genuine abstention from hedged fabrication.

Pattern 2: Template-based fabrication.

Llama-3.3-70B generates near-identical responses across distinct CVEs, defaulting to generic descriptions such as "buffer overflow in a widely used library allowing remote code execution" regardless of the actual vulnerability type. This pattern is consistent with its near-zero judge faithfulness score ($\bar{s} = 0.190$) and lowest standard deviation in H (0.19), confirming uniform fabrication rather than variable-quality reasoning.

Table 4 Model Statistical Performance Summary

Stat	DS	GPT	Gemma	Qwen	Llama
Mean	0.890	0.60	0.611	0.48	0.82
Median	0.754	0.5	0.71	0.682	0.743
Std Dev	0.17	0.2	**0.34**	0.34	0.15
% > 0.5	90.3	44.5	78	64.2	99.6

Pattern 3: Correct category, wrong details.

Models frequently identify the high-level vulnerability class correctly while fabricating specifics affected versions, CVSS scores, and patch status. This partial accuracy accounts for the moderate entailment scores ($\bar{e}$ = 0.34 - 0.5) coexisting with low judge faithfulness, and is the primary reason the Claude judge assigns higher $\bar{s}$ than Qwen3-32B: it awards partial credit where Qwen3-32B treats any fabrication as binary failure.

Pattern 4: Confidence escalation.

Several models open with hedging language but progressively increase assertiveness, concluding with definitive claims unsupported by training data. This within-response drift suggests that autoregressive generation momentum overrides initial uncertainty calibration, a failure mode not captured by response-level classification which assigns a single category to the entire output.

5. Conclusion and Future Work

5.1 Summary of Findings

We presented HalluCVE, a multi-signal benchmark for detecting hallucinations in LLM-generated Cyber Threat Intelligence. Our evaluation of five state-of-the-art LLMs on 1000 CVE entries revealed several critical findings:

1) Pervasive hallucination: All models exhibit substantial hallucination rates (mean Hallucination Index 0.480–0.820), with hallucination-positive rates ranging from 44.5% to 99.6% of samples using a 0.5 threshold.

2) Dangerous confidence: Models frequently provide confident, authoritative-sounding responses for vulnerabilities they cannot possibly know, with confabulation rates reaching 100% for post-cutoff CVEs.

3) Scale does not ensure safety: Larger models do not demonstrate better uncertainty calibration; Qwen2.5-72B achieved the best safety score (0.808) with an 80.75% refusal rate on unknown CVEs, while the comparably-sized Llama-3.3-70B confabulated on 100% of them.

4) Multi-signal detection is essential: Individual detection components capture different failure modes;

their combination provides more comprehensive and robust assessment than any single method.

5) Self-awareness without prevention: Some models accurately assess their own outputs as unfaithful when prompted for self-reflection, yet still generate those fabrications initially, revealing a disconnect between evaluation and generation capabilities.

These findings underscore the urgent need for hallucination mitigation in security-critical LLM applications. Current models are not ready for unsupervisedcdeployment in vulnerability intelligence workflows.

5.2 Why Hallucinations Occur

Understanding the root causes of hallucination is essential for developing effective mitigations:

Training Data Limitations. LLMs are trained on static snapshots of data with inherent cutoff dates. When queried about post-cutoff information, models face a fundamental knowledge gap. Rather than acknowledging this gap, models often extrapolate from related training examples, generating plausible but fabricated content [3]. The pressure to provide helpful responses overrides accurate uncertainty expression.

Optimization for Fluency. LLM training objectives, including next-token prediction and Reinforcement Learning from Human Feedback (RLHF), optimize for fluent, coherent, and helpful responses. This creates implicit pressure to provide substantive answers even when the model lacks relevant knowledge, as refusal may be penalized during training if evaluators interpret it as unhelpful [20].

Pattern Matching Over Reasoning. LLMs excel at pattern matching and statistical continuation but lack robust mechanisms for distinguishing memorized facts from plausible guesses. Models may generate responses that match the form of expected outputs (technical vulnerability descriptions with version numbers, CVSS scores, affected components) without grounding in factual content. The model recognizes "CVE description" as a linguistic pattern and fills in plausible sounding technical details.

Distributional Shift. CTI content exhibits rapid distributional shift as new vulnerabilities, attack techniques, and affected products emerge continuously. Models trained on historical data may not generalize well to novel terminology, software versions, or vulnerability patterns that differ from training distribution [10].

Lack of Uncertainty Calibration. Most LLMs are poorly calibrated for uncertainty expression, lacking mechanisms to reliably distinguish "I know this with high confidence" from "I'm guessing based on weak patterns." This leads to uniformly confident outputs regardless of actual knowledge state [21]. The absence of explicit uncertainty quantification during training exacerbates this problem.

5.3 Strategies for Mitigation

Based on our findings and the literature, we identify several strategies for reducing hallucination in CTI applications:

5.3.1 Retrieval-Augmented Generation (RAG)

RAG systems augment LLM generation with retrieved evidence from authoritative sources [22]. For CTI applications, this means querying the National Vulnerability Database, vendor security advisories, and threat intelligence feeds at inference time, grounding responses in current, verified information. The Crimson system [18] demonstrated that Retrieval-Aware Training significantly reduces hallucination rates in CVE-to-ATT&CK mapping tasks.

Implementation considerations:

- Maintain up-to-date indices of authoritative CTI sources with regular synchronization
- Implement citation requirements that link specific claims to retrieved evidence documents
- Use retrieval confidence scores to modulate response certainty and trigger human review
- Deploy fallback mechanisms that refuse to respond when no relevant documents are retrieved

5.3.2 Uncertainty Quantification and Calibration.

Training models to express calibrated uncertainty is crucial for safety-critical applications [21]. Promising approaches include:

- Verbalized confidence: Training models to preface responses with explicit confidence levels calibrated to actual accuracy
- Abstention training: Explicitly rewarding appropriate refusal behavior during RLHF, treating "I don't know" as a high-quality response when warranted
- Ensemble methods: Using agreement across multiple independent samples or models as a calibrated confidence proxy
- Token probability analysis: Leveraging internal model uncertainty signals like entropy and perplexity to detect low-confidence generations

5.3.3 Domain-Specific Fine-Tuning.

Fine-tuning on curated CTI datasets with explicit hallucination penalties may improve factual grounding. Key considerations:

- Include negative examples explicitly demonstrating appropriate refusal for unknowable queries
- Apply strong penalties to fabrication of specific technical details (CVE IDs, version numbers, CVSS scores, CWE classifications)
- Incorporate fact-verification as an auxiliary training objective, rewarding models for accurate entity extraction and validation
- Balance helpfulness rewards with strict factual accuracy requirements to avoid rewarding plausible fabrication

5.3.4 Post-Generation Verification.

Deploying automated verification systems as a post-processing step can catch hallucinations before they reach end users:

- Entity validation: Verify that mentioned CVE IDs, CWE classifications, product names, and version numbers exist in authoritative databases
- Cross-reference checking: Compare generated claims against NVD records, vendor advisories, and MITRE data sources
- Self-consistency filtering: Generate multiple independent responses and reject those showing high variance, as inconsistency signals hallucination
- HalluCVE integration: Deploy the HalluCVE framework as a real-time verification layer computing hallucination scores before displaying outputs

5.3.5 Human-in-the-Loop Workflows**.**

For highstakes CTI applications, maintaining human oversight remains essential in the near term:

- Present LLM outputs explicitly as drafts requiring analyst verification, not authoritative reports
- Highlight low-confidence claims and unknown entities for prioritized manual review
- Implement feedback loops to capture and correct hallucinations, improving future model iterations
- Establish clear escalation procedures when automated verification flags potential fabrications

5.4 Future Directions

Several research directions emerge from this work:

- Domain-specific NLI models: Training NLI models on cybersecurity corpora to improve entailment detection accuracy for technical content with specialized terminology.
- Fine-grained hallucination taxonomy: Developing CTI-specific hallucination categories (version fabrication, CVSS inflation, CWE misclassification, exploit availability errors) enabling targeted detection and mitigation.
- Real-time SOC integration: Deploying HalluCVE as a verification layer in Security Operations Centers to prevent hallucinated intelligence from propagating through decision pipelines and incident response workflows.
- Mitigation benchmarking: Systematically evaluating the effectiveness of RAG, fine-tuning, uncertainty quantification, and other mitigation strategies using HalluCVE as the standardized evaluation framework.
- Adversarial robustness: Investigating whether adversarial prompting techniques can induce higher hallucination rates and developing defenses against such attacks.
- Longitudinal evaluation: Tracking hallucination rates as models are updated over time to assess whether newer versions exhibit improved calibration and reduced fabrication.

5.5 Conclusion

HalluCVE provides the first dedicated benchmark for evaluating hallucination in LLM-generated Cyber Threat Intelligence, addressing a critical gap in the assessment of AI systems for security applications.

Our findings reveal that current state-of-the-art LLMs pose significant risks when deployed for vulnerability analysis, frequently generating confident fabrications about unknown vulnerabilities with confabulation rates reaching 100%.

These results have important implications for cybersecurity practitioners. LLMs should not be deployed unsupervised for vulnerability intelligence generation. Organizations must implement robust verification mechanisms, maintain human oversight, and treat LLM outputs as drafts requiring validation rather than authoritative assessments.

As LLMs become increasingly integrated into Security Operations Centers and threat intelligence platforms, robust hallucination detection and mitigation will be essential for maintaining the integrity of security decision-making. The HalluCVE framework provides both a benchmark for measuring progress and a practical toolkit for deployment in production environments.

The HalluCVE framework, evaluation dataset, and model outputs are released as open-source resources to support reproducible research and practical adoption in cybersecurity operations. We hope this work catalyzes further research into safe and reliable AI systems for security-critical applications.

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